

IMDb Playwright Automation Framework

Capstone Project Plan

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1. Project Overview

This project is an end-to-end UI automation testing framework built using Playwright with JavaScript, focused on validating IMDb functionalities. The framework follows the Page Object Model (POM) design pattern for maintainability, modularity, and scalability.

2. Project Objectives

Build a structured Playwright automation framework
Implement Page Object Model (POM)
Automate homepage, search, movie details, and navigation flows
Maintain reusable page components
Ensure clean separation of test and page logic
Generate Playwright HTML reports
Capture screenshots/videos on failure
Maintain scalable architecture for future expansion

3. Technology Stack

Technology	Purpose
Playwright	Browser automation framework
JavaScript	Programming language
Node.js	Runtime environment
Git & GitHub	Version control
VS Code	Development environment

4. Core Functional Modules

Home Module – UI validation and elements verification
Search Module – Movie search functionality testing
Movie Module – Movie details and metadata validation
Navigation Module – Routing and navigation testing

5. Framework Features

Page Object Model (POM) architecture
Reusable page classes and utilities
Screenshot capture and video recording support
Trace viewer support for debugging
HTML report generation
Scalable folder structure

6. Project Folder Structure

```
imdb-playwright-project/  
├── tests/  
│   ├── home.spec.js  
│   ├── search.spec.js  
│   ├── movie.spec.js  
│   └── navigation.spec.js  
├── pages/  
│   ├── homePage.js  
│   ├── searchPage.js  
│   └── moviePage.js  
├── utils/  
│   ├── testData.js  
│   └── helpers.js  
├── playwright.config.js  
├── package.json  
└── README.md
```

7. Execution Roadmap

Day	Task
Day 1	Setup Playwright project + folder structure
Day 2	Create Page Object Model (Home + Base structure)
Day 3	Implement Search functionality automation
Day 4	Movie details validation module
Day 5	Navigation test cases
Day 6	Add utilities + debugging
Day 7	Reporting setup + execution stability
Day 8	Final testing + optimization

8. Expected Outcomes

A working Playwright automation framework Strong understanding of Page Object Model (POM) Ability to automate real-world UI workflows Modular test design structure Debugging using Playwright trace viewer A portfolio-ready QA automation project

9. Conclusion

Conclusion: This capstone project demonstrates industry-level automation testing practices using Playwright. The framework is scalable, maintainable, and suitable for future enhancements.